

Casavista Plugin — Designer Guide

For designers who already use 3ds Max. This guide covers **only what you need for the Casavista modifier** so your scene exports correctly to Unreal (Datasmith/Dataprep): interactables can swap models, materials can target the right part of a mesh, vegetation is tagged, and doors open from the correct hinge.

What the plugin does

Casavista is a **modifier** that tags objects for the Unreal pipeline. You add it, set **Class**, and where needed fill **Models** or **Materials** and set **pivots**. The rest is standard Max; this guide only covers the Casavista-specific choices.

Adding the Casavista modifier

Add **Casavista Property Editor** from the Modifier List (it may appear under a category like "DCG Samples"). The modifier adds a rollout with **Class**, **Models**, and **Materials**. See Figure 1.

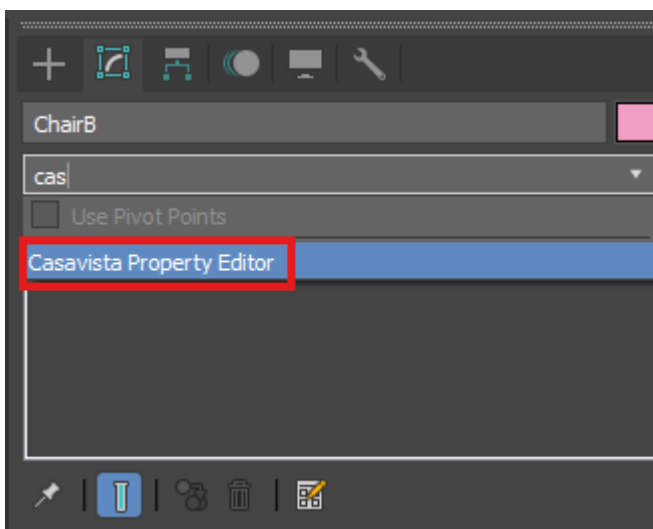


Figure 1. Casavista Property Editor in the Modifier List.

The Casavista rollout

Three areas control how the object is tagged:

1. Class

Class dropdown: **None**, **Interactable**, **Vegetation**, **Door**. Pick the one that matches the object. **None** is only a placeholder—if you don't need the modifier on that object, remove the modifier instead of leaving Class on None. See Figure 2.

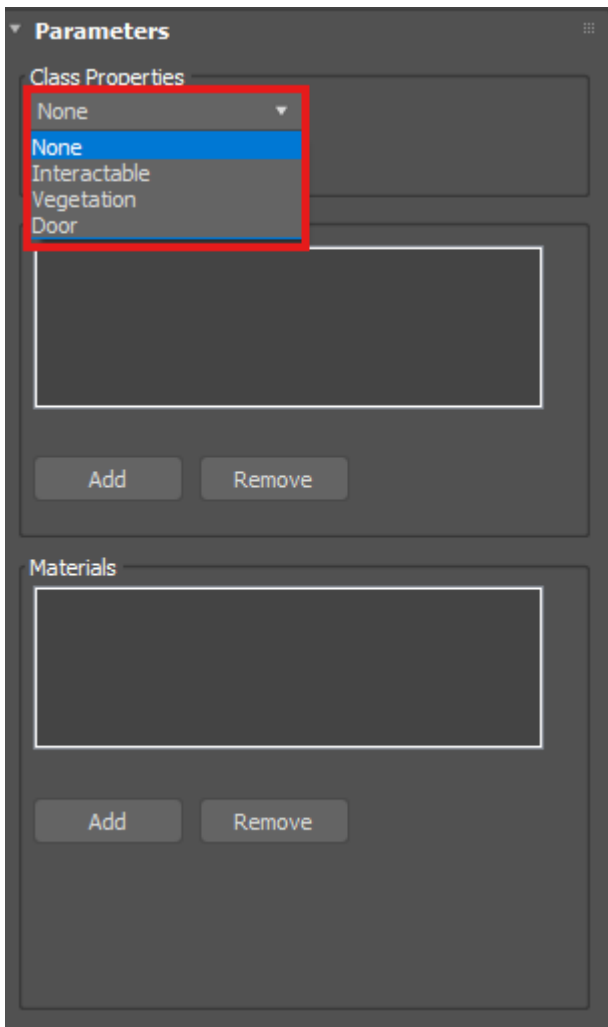


Figure 2. Class options.

2. Models

A list plus **Add / Remove**. Use it when the object can **swap to other mesh variants** (e.g. one chair slot that can be Chair A, B, or C). The object that has the modifier is the **main** one; add **only the other variants** to the list (e.g. Chair_B, Chair_C—not the main chair). See Figure 3.

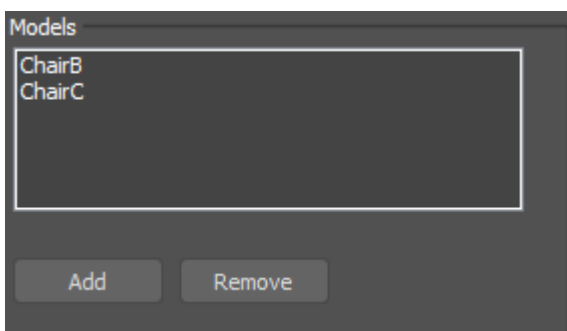


Figure 3. Models list and Add / Remove.

3. Materials

A list plus **Add / Remove**. Use it when the **shape stays the same** but you want to drive **specific parts** (e.g. only the tabletop). You pick a material and a **Material Index** (0, 1, 2, ...) so the pipeline knows which part to change. See Figure 4.

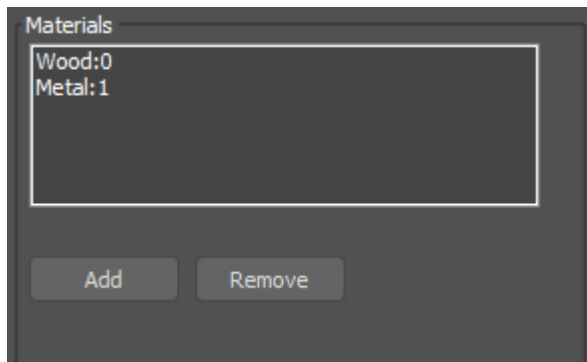


Figure 4. Materials list and Add / Remove.

Adding models (Models list)

When: Class = **Interactable** and you want alternative mesh options.

- **Add** opens the **Model Select** dialog: search box at top, scene objects below. Use the search to filter, select the **alternative** meshes only (not the object that has the modifier), then **OK**. Names appear in the Models list; **Remove** takes them off. See Figure 19.

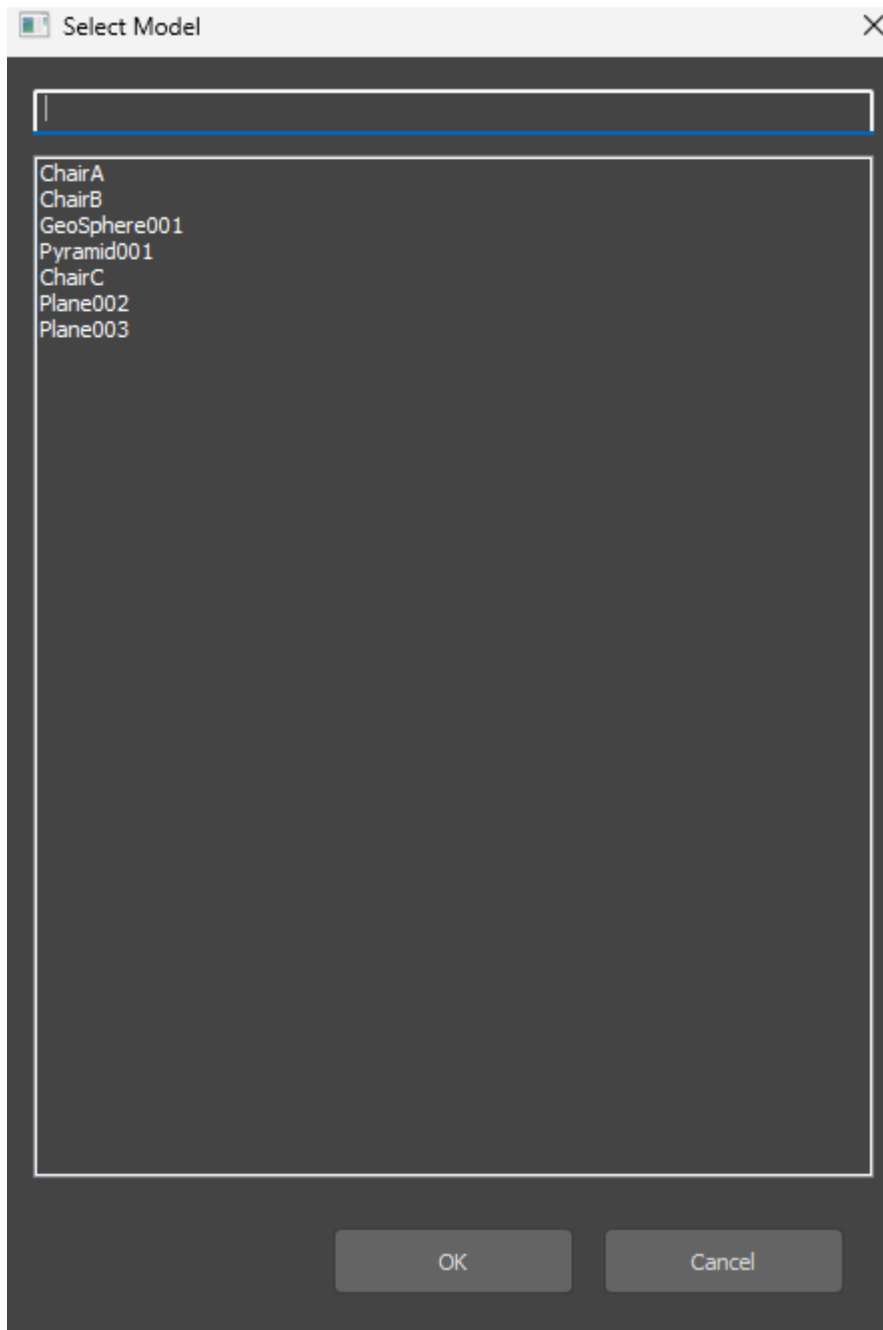


Figure 19. Model Select dialog.

Adding materials (Materials list)

When: Same shape, but you want to drive **specific parts** (e.g. tabletop only).

- **Add** opens the **Material Select** dialog: pick a material from the list (search if needed), set **Material Index** (0, 1, 2, ...) for the part you want, then **OK**. Entry shows as e.g. **Wood_Material (Index: 0)**. **Remove** takes entries off. You need the correct index for each part (see "Material Index" below). See Figure 20.

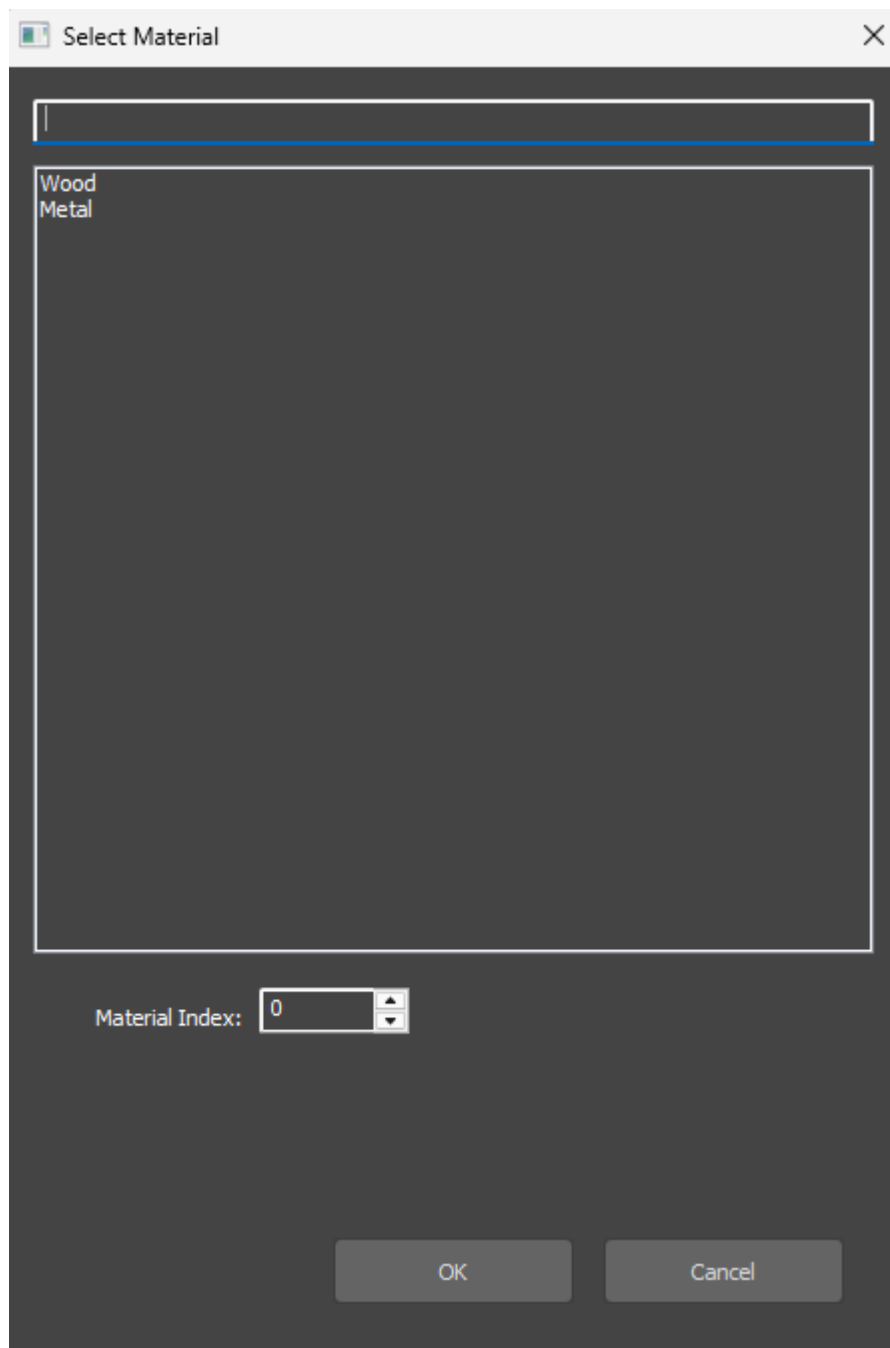


Figure 20. Material Select dialog.

Class: what to set and when

None

None = no class chosen yet. If the object doesn't need Casavista at all (no swap, no material drive, not vegetation, not a door), **remove the modifier** instead of leaving Class on None. See Figure 21.

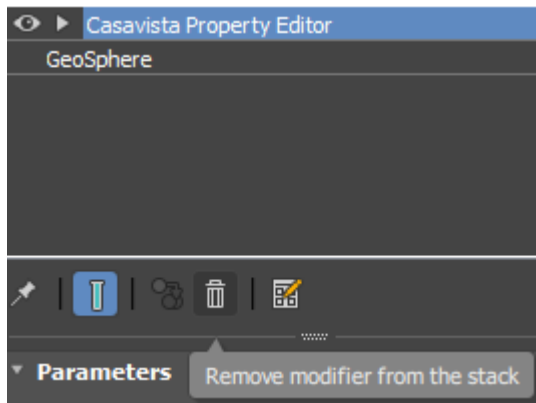


Figure 21. Removing the Casavista modifier.

Interactive

Use for: Objects that can **swap to other mesh variants** in Unreal (e.g. one chair slot that can be Chair A, B, or C). The object with the modifier is the **main**; the Models list must contain **only the other variants**, not the main one.

- Add Casavista → **Class** = **Interactive** → **Add** under Models → in Model Select, pick only the alternative meshes → OK.
- **Pivots:** Main and every mesh in the Models list must share the **same pivot position** (e.g. seat center or base). If they don't, the swap will be misaligned in Unreal. Align with **Affect Pivot Only** (see "Pivots" below). See Figures 5–6, 10–11.

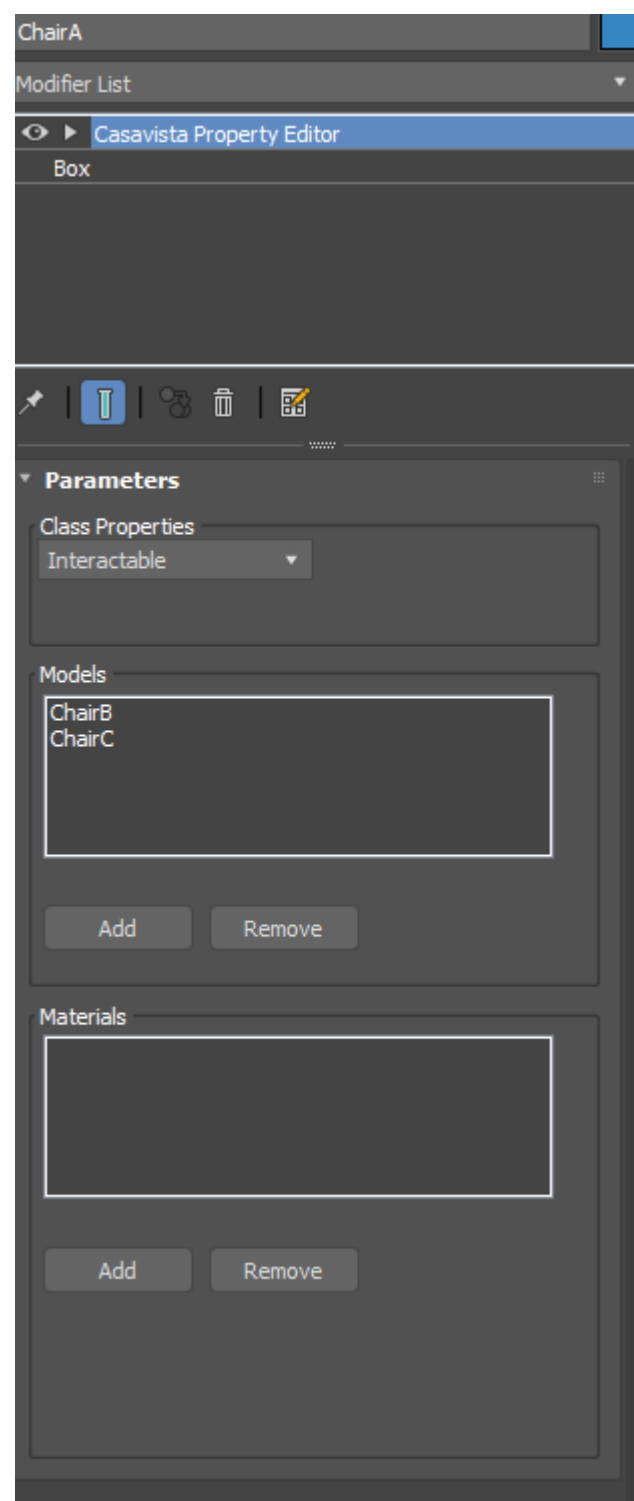


Figure 5. Interactable: main object + Models list with alternatives only.

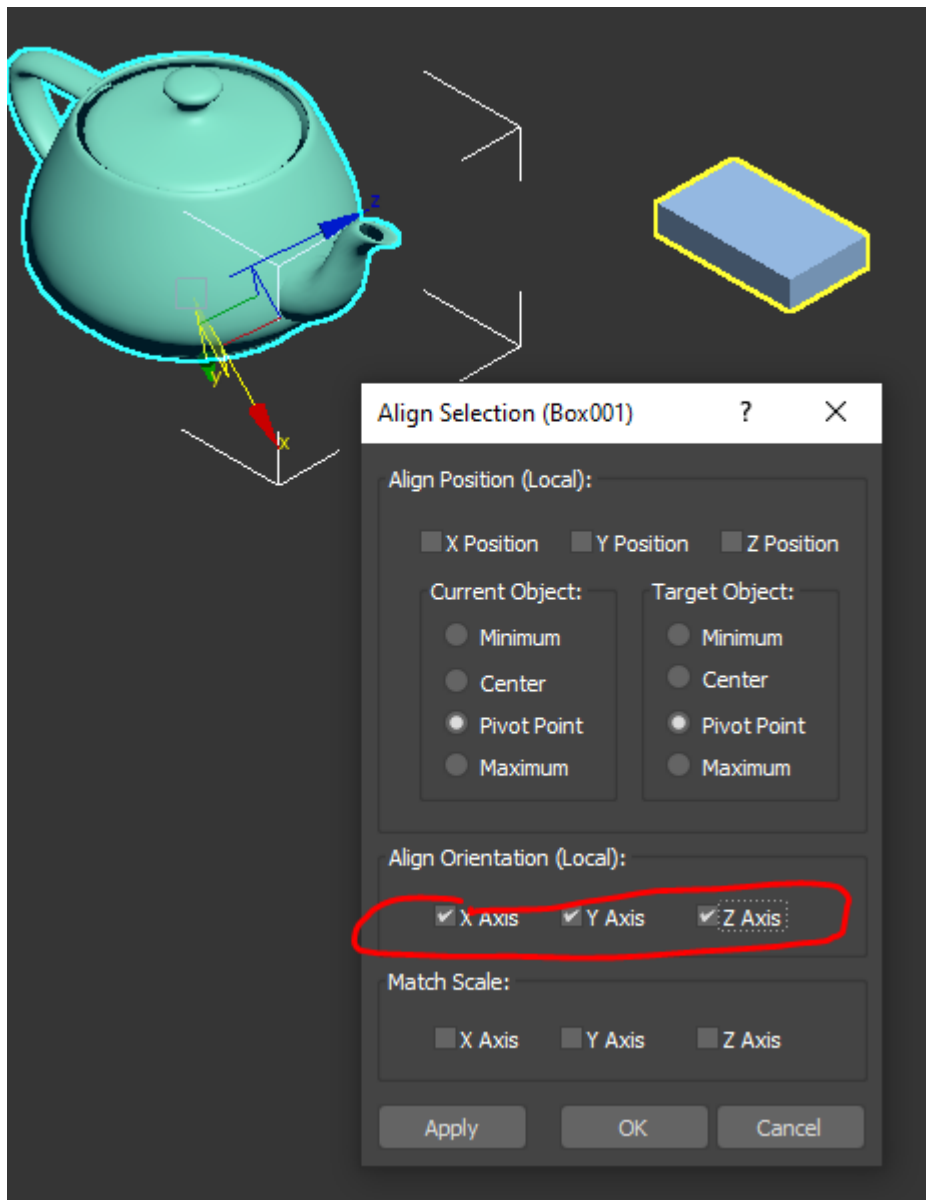


Figure 6. Same pivot.

Vegetation

Use for: Plants, trees, bushes, grass, etc. Add Casavista → **Class = Vegetation**. No Models or Materials needed. Use **real geometry** (no proxies). See Figure 7.

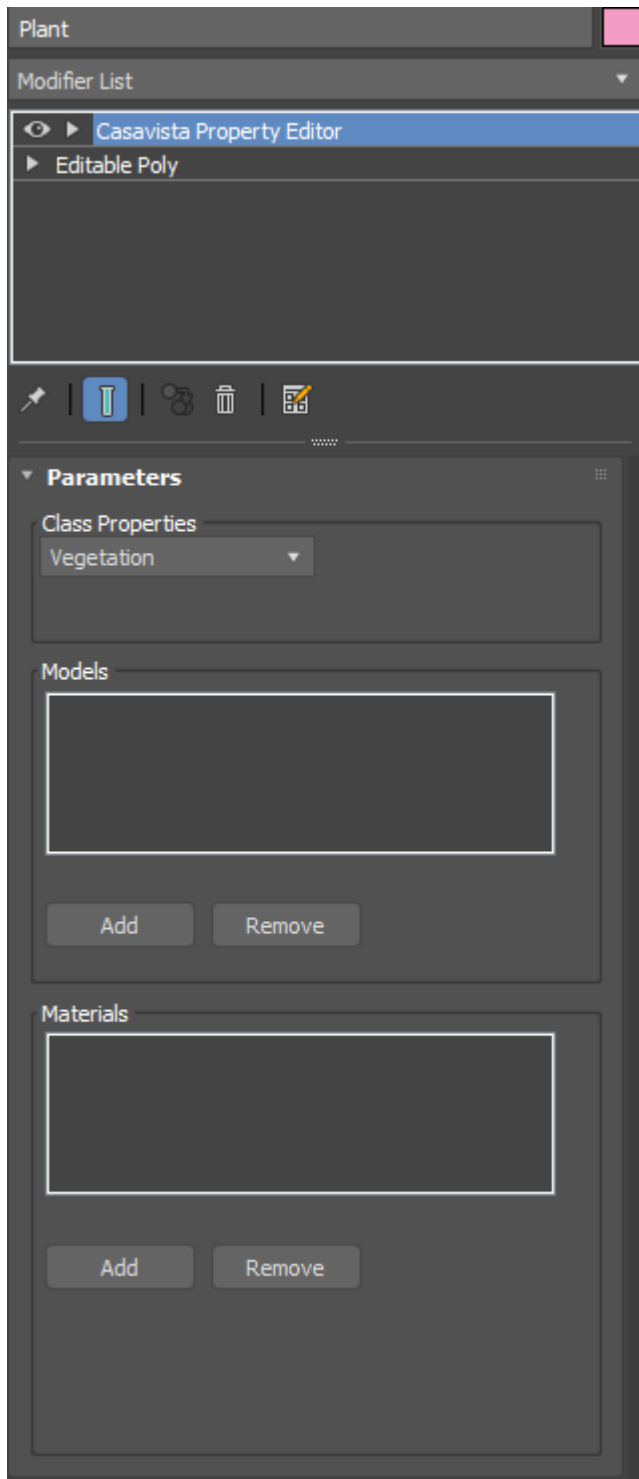


Figure 7. Vegetation: Class = Vegetation only.

Door

Use for: Doors that open/close in Unreal. The door must be **one mesh** (panel + frame + handle merged). Add Casavista → **Class = Door**. **Pivot** must be on the **hinge edge** (the edge that doesn't move when the door opens). Set it with **Affect Pivot Only** (see "Pivots" below).

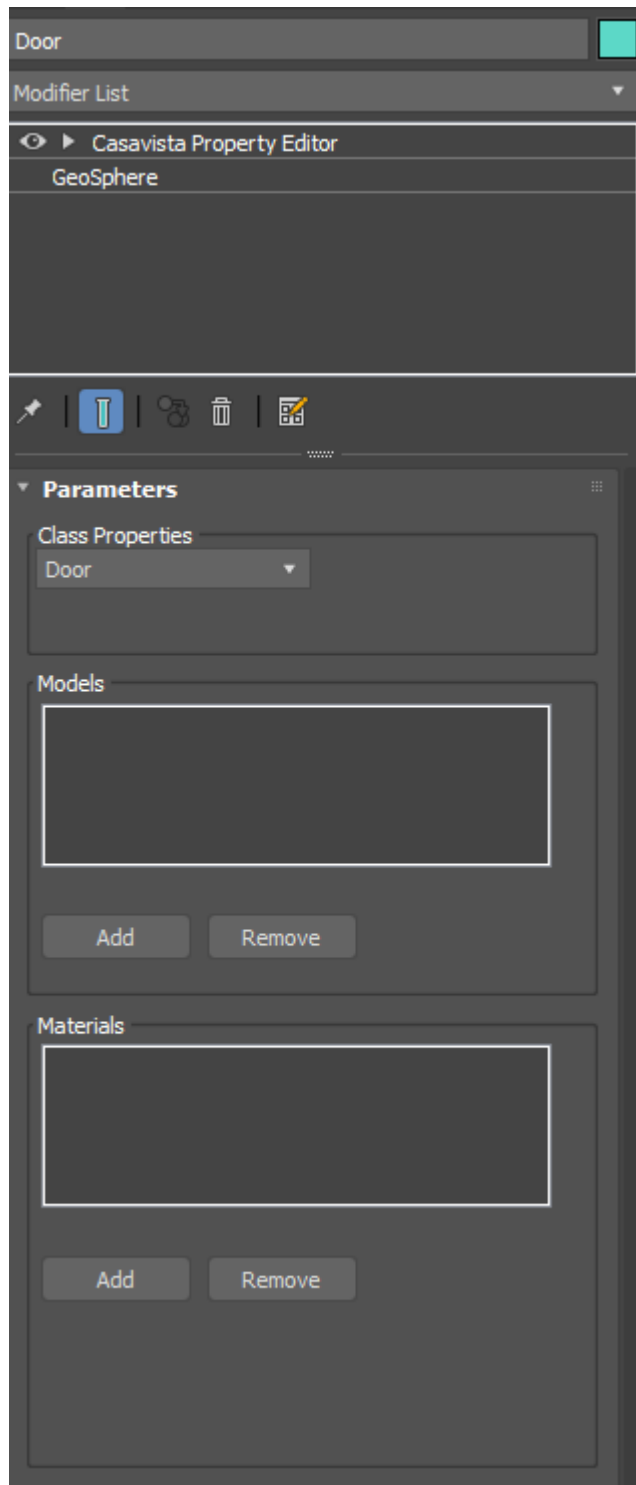


Figure 8. Door: single mesh, Class = Door.

Pivots (Casavista rules)

- **Interactable:** Main object and every mesh in the Models list must have the **same pivot position** (e.g. seat center or base) so swaps line up in Unreal.
- **Door:** Pivot must be on the **hinge edge** (the edge that stays fixed when the door opens). Rotate in Max to confirm it swings correctly.

How: Hierarchy → **Affect Pivot Only** → move pivot (or Align to main/door edge) → turn **Affect Pivot Only** off.

Material Index

What it is: Multi/Sub-Object (or similar) slots are numbered 0, 1, 2, ... In Casavista Materials you pick a material and an **index** so the pipeline knows which part of the mesh to change (e.g. 0 = tabletop, 1 = legs).

Finding the index: Sub-material order in the mat editor = index order (first = 0, second = 1, ...). If unsure, give each sub-material a different color and check which part on the model is which; use that number in Material Select.

Quick reference

Class	Use for	Models list	Materials list	Pivot / geometry
None	Placeholder only; if you don't need the modifier, remove it	—	—	—
Interactable	Objects that swap model variants	Add only the alternative options (not the main object)	Optional	Align pivots of all alternatives with main object
Vegetation	Plants, trees, bushes	Leave empty	Leave empty	Plain geometry, no proxies
Door	Doors that open/close	Leave empty	Optional	Single merged mesh; pivot on hinge edge